

ACTIVE DIRECTORY HOME LAB

Phase 1 — Environment Build & IT Administration

Analyst: Bryan Sykes | May 27, 2026 | IT Administration & Security Portfolio Project

1. Project Metadata

Project Title	Active Directory Home Lab Phase 1: Environment Build & IT Administration
Phase	Phase 1 of 2 (Phase 2: Attack & Detection coming separately)
Date	May 27, 2026
Analyst	Bryan Sykes
Domain	apexdefense.local
Domain Controller	DC01 Windows Server 2022 Standard Evaluation
Virtualization	Oracle VirtualBox Personal PC (32GB DDR5, PCIe 3.0 SSD)
Host OS	Windows (Personal PC)
Phase 1 Coverage	AD DS Installation, Domain Controller Promotion, OU Structure, Users, Groups, GPOs, Help Desk Scenarios
Phase 2 Preview	Windows 10 Workstation, Kali Linux Attacker, Kerberoasting, Pass-the-Hash, BloodHound, LLMNR Poisoning, Event Log Detection

2. Project Overview

This project builds a fully functional Active Directory environment from scratch in a personal VirtualBox lab, then uses it as the target for simulated enterprise attack techniques in Phase 2. The project spans two skill domains: IT administration and security operations deliberately designed to demonstrate that both capabilities are present in a single, cohesive lab environment.

Phase 1 covers the build and administration layer. Windows Server 2022 was installed, promoted to a Domain Controller, and configured with a production-realistic Active Directory structure including Organizational Units, domain users, security groups, and Group Policy Objects. Five help desk scenarios were simulated against the live domain to demonstrate day-to-day IT support operations in an AD environment.

The fictional organization Apex Defense Solutions is the same government contractor used across this portfolio, creating a consistent narrative thread that connects the Active Directory lab to the previously completed Vendor Risk Assessment and other portfolio projects.

3. Lab Environment

Component	Details
Host Machine	Windows OS, 32GB DDR5 RAM, PCIe 3.0 SSD (789GB free)
Virtualization Platform	Oracle VirtualBox
Domain Controller VM	DC01 Windows Server 2022 Standard Evaluation (Desktop Experience)
DC01 Resources	4096MB RAM, 2 vCPUs, 60GB VDI storage
Network Configuration	VirtualBox Internal Network apexdefense-lab (isolated, no internet)
DC01 IP Address	192.168.1.10 (static)
DC01 DNS	127.0.0.1 (self-referencing required for AD-integrated DNS)
Domain Name	apexdefense.local
NetBIOS Name	APEXDEFENSE
Forest/Domain Functional Level	Windows Server 2016
Phase 2 VMs (Planned)	Windows 10 Workstation + Kali Linux Attacker configured in Phase 2

4. Windows Server 2022 Installation

4.1 ISO Acquisition

Windows Server 2022 evaluation ISO was obtained from Microsoft Evaluation Center (microsoft.com/en-us/evalcenter/evaluate-windows-server-2022). Registration with basic contact information is required. The evaluation license provides 180 days of full functionality at no cost standard practice for home lab environments. Downloaded file: SERVER_EVAL_x64FRE_en-us.iso (4.70GB).

4.2 VM Configuration

VirtualBox VM created with the following configuration before installation:

- Name: DC01-ApexDefense
- RAM: 4096MB (increased from default 2048MB required for stable AD DS operation)
- vCPUs: 2
- Storage: 60GB VDI on PCIe 3.0 SSD
- Network: Internal Network (apexdefense-lab) changed from default NAT to enable VM-to-VM communication
- ISO: SERVER_EVAL_x64FRE_en-us.iso attached to optical drive

4.3 Installation Process

Windows Server 2022 Standard Evaluation (Desktop Experience) selected the GUI variant providing full Server Manager and administrative tools. Custom installation type selected to perform a clean install on the 60GB virtual disk.

Note: The installation required a manual boot from ISO on second attempt. The initial boot attempt produced a black screen because the VM tried to boot from the empty hard disk rather than the ISO. Issue resolved by resetting the VM and pressing a key at the 'Press any key to boot from CD or DVD' prompt within the 2-second window.

A third partition screen appeared during the second installation attempt showing existing partitions from the first incomplete install attempt. Windows offered to overwrite the previous installation (creating a Windows.old folder). Confirmed and proceeded installation completed successfully on this attempt.

4.4 Post-Installation Configuration

After installation and initial login as local Administrator, two configuration steps were completed before Active Directory installation:

Server Rename

Default computer name (randomly generated) renamed to DC01 via Server Manager → Local Server → Computer Name → Change. Restart deferred until after static IP configuration.

Static IP Address

Network adapter configured with a static IP via Network Connections → IPv4 Properties:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Preferred DNS: 127.0.0.1 self-referencing DNS required for Active Directory domain resolution

The adapter shows 'Unidentified Network' status expected behavior for an Internal Network adapter in VirtualBox with no upstream DHCP or router. Functional for AD purposes.

Server restarted to apply the name change. Post-restart Server Manager confirmed computer name: DC01.

5. Active Directory Domain Services Installation & Promotion

5.1 AD DS Role Installation

Active Directory Domain Services role installed via Server Manager → Add Roles and Features Wizard:

- Installation type: Role-based or feature-based installation
- Server: DC01 (local server)
- Role selected: Active Directory Domain Services
- Required features accepted when prompted (AD DS dependencies)
- Installation completed successfully — confirmed by 'Installation succeeded on DC01' message

5.2 Domain Controller Promotion

After AD DS role installation, the yellow flag notification in Server Manager indicated the server was ready for promotion. Clicked 'Promote this server to a domain controller' to launch the AD DS Configuration Wizard.

Configuration selections:

- Deployment operation: Add a new forest (first DC in a brand new domain)
- Root domain name: apexdefense.local
- Forest functional level: Windows Server 2016
- Domain functional level: Windows Server 2016
- DNS Server: Enabled (AD-integrated DNS installed automatically)
- Global Catalog: Enabled
- DSRM password: Set (same as Administrator password for lab simplicity)
- NetBIOS domain name: APEXDEFENSE (auto-generated, accepted)
- Database, Log, SYSVOL paths: Default locations accepted

DNS delegation warning appeared and was dismissed as expected behavior when creating a new forest with no parent DNS zone. Prerequisite check passed successfully. Installation initiated and server restarted automatically upon completion.

Post-restart login screen confirmed successful promotion: login showed APEXDEFENSE\Administrator instead of local Administrator confirming the domain is active and DC01 is operating as the Domain Controller for apexdefense.local.

Server Manager post-promotion shows two new entries in the left sidebar: AD DS and DNS both automatically added when the DC role was activated.

6. Active Directory Structure

6.1 Organizational Units

A parent OU named _DEPARTMENTS was created directly under apexdefense.local. The underscore prefix is a common enterprise convention that sorts administrative OUs

alphabetically above default AD containers for easier navigation. Four department OUs were created inside _DEPARTMENTS:

Organizational Unit	Purpose
_DEPARTMENTS	Parent container for all department OUs direct child of apexdefense.local
_DEPARTMENTS\IT	IT department users and IT-Admins security group
_DEPARTMENTS\HR	Human Resources users, HR-Staff group, Desktop Restrictions GPO linked
_DEPARTMENTS\Finance	Finance users, Finance-Team group, Desktop Restrictions GPO linked
_DEPARTMENTS\Management	Management users and Management-Team group
_DISABLED_ACCOUNTS	Holding OU for disabled terminated employee accounts direct child of apexdefense.local

'Protect container from accidental deletion' was enabled on all OUs — best practice in production environments to prevent inadvertent OU deletion which would disassociate all contained objects from their GPO links.

6.2 Domain Users

Full Name	Username	Department OU	Security Group	Status
John Carter	jcarter	IT	IT-Admins	Active
Sarah Mitchell	smitchell	IT	IT-Admins	Active
James Reynolds	jreynolds	HR	HR-Staff	Active
Lisa Thompson	lthompson	HR	HR-Staff	Active
Michael Torres	mtorres	Finance	Finance-Team	Active
Angela Brooks	abrooks	Finance	Finance-Team	Active
David Hawkins	dhawkins	Management	Management-Team , Finance-Team	Active
Rachel Kim	rkim	Management	Management-Team	Active
Kevin Park	kpark	_DISABLED_ACCOUNTS	IT-Admins (membership retained)	Disabled — Terminated

6.3 Security Groups

Group Name	Scope	Type	Members
IT-Admins	Global	Security	jcarter, smitchell, kpark (disabled)

HR-Staff	Global	Security	jreynolds, lthompson
Finance-Team	Global	Security	mtorres, abrooks, dhawkins
Management-Team	Global	Security	dhawkins, rkim

7. Group Policy Configuration

Three Group Policy Objects were created and linked to control security settings and user environment restrictions across the domain and specific OUs.

7.1 Apex-Password-Policy (Domain-Wide)

Linked to apexdefense.local — applies to all users in the domain. Configured under Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy:

Setting	Configured Value	Purpose
Enforce password history	10 passwords remembered	Prevents reuse of recent passwords
Maximum password age	90 days	Forces regular password rotation
Minimum password age	1 day	Prevents immediate re-use after change
Minimum password length	12 characters	Increases brute force resistance
Password complexity requirements	Enabled	Requires uppercase, lowercase, number, special character

7.2 Apex-Password-Policy — Account Lockout (Domain-Wide)

Configured under the same GPO, Account Policies → Account Lockout Policy:

Setting	Configured Value	Purpose
Account lockout threshold	5 invalid logon attempts	Triggers lockout after 5 consecutive failures
Account lockout duration	30 minutes	Auto-unlocks after 30 minutes without admin intervention
Reset account lockout counter	30 minutes	Resets failed attempt count after 30 minutes of no failures

7.3 Apex-Desktop-Restrictions (HR and Finance OUs)

Linked to both HR and Finance OUs — applies only to users in those departments. Configured under User Configuration → Policies → Administrative Templates:

Setting	Value	Business Justification
Prevent changing desktop background	Enabled	Enforces corporate branding and professional appearance
Prevent changing screen saver	Enabled	Ensures screen lock policy cannot be circumvented by users
Prevent access to command prompt	Enabled	Reduces attack surface. Standard users should not need CMD access
Remove Run menu from Start Menu	Enabled	Prevents users from launching unauthorized executables directly

Note: During GPO creation a typo was introduced in the password policy GPO name (Apex-Passowrd-Policy instead of Apex-Password-Policy). The GPO functions correctly; the name is cosmetic only. Documented here for transparency and as a reminder that GPO names should be verified before deployment in production environments.

8. Help Desk Scenarios

Five help desk scenarios were simulated against the live apexdefense.local domain to demonstrate day-to-day IT support operations in an Active Directory environment. Each scenario represents a common real-world ticket type that a help desk analyst would encounter.

Scenario 1 — New User Onboarding

Situation	A new employee, Kevin Park, joined the IT department and requires a domain account with appropriate group membership.
Screenshot	Screenshot 16 - 16-helpdesk-new-user-onboarding.png)

Steps performed:

- Navigated to **_DEPARTMENTS > IT OU** in Active Directory Users and Computers
- Created new user account: Kevin Park (kpark) with standard lab password
- Set Password never expires for lab environment (noted as production anti-pattern)
- Opened IT-Admins group properties → Members tab → Added kpark
- Verified Kevin Park appeared in IT-Admins membership list

Outcome: Account created in correct OU and added to IT-Admins security group successfully. Performed from memory without step-by-step guidance.

Scenario 2 — Password Reset

Situation	Lisa Thompson (lthompson) forgot her password and requires a reset with forced change at next logon.
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Screenshot

Screenshot 17 — 17-helpdesk-password-reset.png

Steps performed:

- Located lthompson in HR OU
- Discovered 'User must change password at next logon' was grayed out caused by 'Password never expires' conflict
- Opened lthompson Properties → Account tab → Unchecked 'Password never expires'
- Right-clicked lthompson → Reset Password → Set TempPass@456
- Checked 'User must change password at next logon' and 'Unlock the user's account'

Outcome: Password reset successfully with forced change at next logon. Identifying and resolving the Password never expires, conflict documented as a key lesson.

Scenario 3 — Account Lockout Investigation**Situation**

Michael Torres (mtorres) reports he cannot log in. Status unknown may be locked out or disabled.

Screenshot

Screenshot 18 — 18-helpdesk-account-lockout-resolution.png

Steps performed:

- Located mtorres in Finance OU in Active Directory Users and Computers
- Right-clicked → Properties → Account tab to investigate account status
- Determined account was not disabled (no downward arrow on icon)
- Identified 'Password never expires' setting confirmed account configuration
- Performed same remediation as Scenario 2: unchecked Password never expires, reset password, checked both lockout and forced change options

Outcome: Account status investigated via Account tab the correct diagnostic location in ADUC. Remediation applied. Key lesson: Password never expires and Account is locked out are separate flags requiring separate investigation.

Scenario 4 — Permissions Change — Cross-Department Group Membership**Situation**

David Hawkins (dhawkins) was promoted and now requires Finance team resource access in addition to his Management role.

Screenshot

Screenshot 19 — 19-helpdesk-permissions-change.png

Steps performed:

- Located Finance-Team group in Finance OU
- Opened group Properties → Members tab → Add
- Typed dhawkins → Check Names → Resolved to David Hawkins

- Confirmed dhawkins added to Finance-Team alongside existing Finance members

Outcome: dhawkins added to Finance-Team while remaining in the Management OU. Demonstrates the separation between OU location (where a user lives) and group membership (what they can access) a fundamental AD concept.

Scenario 5 — Terminated Employee — Account Disable and Move

Situation	HR issued a termination notice for Kevin Park (kpark). The account must be disabled immediately and moved to a holding OU.
Screenshot	Screenshot 20 — 20-helpdesk-terminated-employee.png

Steps performed:

- Created new OU `_DISABLED_ACCOUNTS` directly under `apexdefense.local` (not inside `_DEPARTMENTS`)
- Located `kpark` in IT OU
- Right-clicked → Disable Account account icon displayed downward arrow confirming disabled status
- Right-clicked `kpark` → Move → Selected `_DISABLED_ACCOUNTS` OU
- Verified Kevin Park appeared in `_DISABLED_ACCOUNTS` with disabled account icon

Outcome: Account disabled and moved to holding OU in a single workflow. This mirrors real enterprise termination procedure: disable immediately (same day as termination), retain in holding OU for 90 days for audit purposes, then permanently delete. Preserving the account temporarily allows recovery if termination is reversed and maintains an audit trail.

9. Key Lessons and Technical Notes

- Password never expires conflicts with User must change password at next logon these settings cannot coexist. In a production environment, standard user accounts should never have Password never expires enabled. Only specific service accounts with managed credentials should use this setting.
- OU location and group membership are independent concepts. A user's OU determines where they live in the directory and which GPOs apply to them by inheritance. Group membership determines what resources they can access. A user can be a member of groups from any OU regardless of their home OU.
- The underscore prefix convention (`_DEPARTMENTS`, `_DISABLED_ACCOUNTS`) sorts administrative OUs above default AD containers alphabetically, making the directory easier to navigate in large environments. This is a widely used real-world convention.
- Protect container from accidental deletion should always be enabled on OUs in production. Accidental OU deletion disassociates all contained objects from their GPO links and can cause widespread policy failures.

- Domain-wide GPOs (linked to the domain root) apply to all users and computers. OU-level GPOs apply only to objects within that OU. GPO inheritance flows downward child OUs inherit policies from parent OUs unless inheritance is blocked.
- DNS self-reference (127.0.0.1) on the Domain Controller is required for Active Directory. The DC must use itself as the primary DNS server because AD relies on AD-integrated DNS for service record resolution. Pointing the DC at an external DNS server breaks domain functionality.
- The SYSVOL share is replicated across all Domain Controllers and stores Group Policy Objects and login scripts. In a single-DC lab it is less visible, but in multi-DC production environments SYSVOL replication health is critical to GPO distribution.
- Terminated employee accounts should be disabled and moved to a holding OU rather than deleted immediately. The standard retention period is 90 days to allow for reversal of termination, ongoing investigations, or audit requirements before permanent deletion.

10. Phase 2 Preview

Phase 2 will build on this environment by adding a Windows 10 domain-joined workstation and a Kali Linux attacker VM, then executing common Active Directory attack techniques against the apexdefense.local domain. Each attack will be documented with exploitation steps, what it looks like in Windows Event Viewer logs, and the detection/prevention controls.

Attack Technique	Description
LLMNR Poisoning	Intercept LLMNR/NBT-NS broadcast requests to capture NTLMv2 password hashes using Responder
Kerberoasting	Request service tickets for service accounts and crack them offline to reveal plaintext passwords
Pass-the-Hash	Authenticate to network resources using a captured NTLM hash without knowing the plaintext password
BloodHound Enumeration	Map the entire AD environment to identify attack paths from low-privilege user to Domain Admin

11. Conclusion

Phase 1 of the Active Directory Home Lab successfully built a fully functional enterprise-grade AD environment from scratch on a personal VirtualBox setup. Windows Server 2022 was installed, promoted to a Domain Controller, and configured with a realistic organizational structure including 4 department OUs, 9 user accounts, 4 security groups, and 3 Group Policy Objects.

Five help desk scenarios were executed against the live domain new user onboarding, password reset, account lockout investigation, cross-department permissions change, and terminated employee handling. Each scenario mirrors a common real-world IT support ticket

and demonstrates practical Active Directory administration skills that directly apply to help desk and junior sysadmin roles.

Phase 2 will elevate this environment from an IT administration lab to a security operations lab by introducing attack simulation and detection, demonstrating that the same AD environment used for day-to-day IT support is also the primary attack target in enterprise intrusions.

Documentation prepared by: **Bryan Sykes** | May 27, 2026 | Active Directory Home Lab — Phase 1